Is whole-body hydration an important consideration in dry eye?

PURPOSE: To identify if whole-body hydration plays an important role in dry eye (DE). We hypothesized that individuals classified as DE have higher plasma osmolality (Posm), indicating suboptimal hydration, compared with those classified as non-DE. METHODS: Using a hospital-based observational cross-sectional design, assessment of DE and hydration was performed upon admission in 111 participants (N = 56 males and 55 females; mean ± SD age 77 ± 8 years). Assessments of DE included tear osmolarity (Tosm), the 5-item dry eye questionnaire (DEQ-5), rating of eye dryness using a visual analogue scale (VAS), and noninvasive tear film breakup time (NITBUT). Hydration assessment was performed by measuring Posm using freezing-point depression osmometry. RESULTS: Posm was higher in DE than control (CON), indicating suboptimal hydration when using the 316 mOsm/L Tosm cutoff for DE (mean Posm + 11 mOsm/kg versus CON, $P = 0.004$, Cohen's effect size [d]) = 0.83) and the more conservative Tosm classification for DE where Tosm >324 and CON <308 mOsm/L (mean Posm + 12 mOsm/kg versus CON, $P = 0.006$, $d = 0.94$). Posm was also higher in DE than CON when using composite DE assessments, including Tosm and DEQ-5 ($P = 0.021$, $d = 1.07$); Tosm and NITBUT ($P = 0.013$, $d = 1.08$); and the VAS and DEQ-5 ($P = 0.034$, $d = 0.58$). CONCLUSIONS: These are the first published data to show that individuals classified as DE have higher Posm, indicating suboptimal hydration, compared with non-DE. These findings indicate that whole-body hydration is an important consideration in DE.